Toll Management System (FASTag)

Problem Statement

Background

A highway company runs toll plazas where vehicles pay toll electronically using FASTag. You are required to design and build a system that manages toll plazas and lanes, vehicles and FASTag accounts, toll processing, rates and passes, and payments and reports. The system must follow the business rules described in each module below.

Module 1 — Toll Plaza and Lane Setup

- Only an Admin can create toll plazas and lanes.
- A toll plaza has a name, a highway name, a location and at least 2 lanes.
- A lane has a number, a type (FASTag, Cash, or Emergency) and a status (Open or Closed). A lane can be closed only if no vehicle is being processed in it.
- A toll rate is set for each plaza, each vehicle type and each trip type (Single or Return), with an effective date. A plaza cannot open until a rate exists for every vehicle type.
- A rate change applies only from its effective date. Earlier transactions are not affected.

Module 2 — Vehicle and FASTag Accounts

- A vehicle has a unique registration number, a type (Car, Van, Bus, Truck, or Multi-axle), an owner name and a phone number.
- A FASTag account has a unique tag ID and is linked to exactly one vehicle. A vehicle can have only one active tag.
- Account status is Active, Blocked or Closed. The account is blocked automatically when the balance falls below Rs. 100.
- A recharge must be at least Rs. 100 and at most Rs. 10,000, and the balance cannot go above Rs. 50,000. The balance can never be negative.
- A blocked account becomes Active again automatically when a recharge takes the balance to Rs. 100 or more.
- An Admin can blacklist a tag (for example, a stolen or fraudulent tag). A blacklisted tag is refused at every plaza.

Module 3 — Toll Processing

- When a vehicle reaches a FASTag lane, the system reads the tag and finds the linked vehicle. The barrier opens only after the toll is deducted successfully.
- If the observed vehicle type does not match the registered type, the higher toll is charged and the transaction is flagged for review.
- A vehicle without a valid Active tag is sent to a Cash lane and pays double the toll.
- If the tag balance is not enough, the barrier does not open. The vehicle is sent to a Cash lane and pays the normal toll in cash.
- Every crossing creates a transaction record with a unique ID, the vehicle, the plaza, the lane, the time, the amount and the status (Success, Failed, or Pending).
- The same tag cannot be charged twice at the same plaza within 2 minutes.
- Emergency vehicles (ambulance, fire engine) use the Emergency lane and pay nothing.

Module 4 — Rates, Discounts and Passes

- If a vehicle returns through the same plaza within 24 hours of a Single trip, the Return trip rate (50% of the Single rate) is applied automatically.
- A vehicle can have a monthly pass for a plaza. A pass has the vehicle, the plaza, a start date and an end date. A vehicle with a valid pass pays nothing at that plaza.
- A vehicle can have only one valid pass per plaza at a time. The pass fee is set by the Admin for each vehicle type.
- The toll amount is rounded to the nearest Rs. 5.

Module 5 — Payments, Disputes and Reports

- Recharge methods are UPI, Card and Net Banking. A recharge record must store the amount, date, method and status (Paid, Failed, or Refunded).
- Cash toll payments are recorded with the amount and the lane operator who collected them.
- A vehicle owner can dispute a wrong deduction within 7 days. If the Admin approves, the amount is refunded to the tag balance.
- When the balance falls below Rs. 200 after a deduction, a low balance alert is generated.
- An Admin can view the daily revenue for each plaza and the number of vehicles for each lane.

SQL Exercise

Based on the system described above, write SQL queries for the following:

Query 1

For each vehicle type, find the total number of toll transactions, the total toll collected, and the average toll per crossing. Show only vehicle types that have more than 2 successful transactions. Sort by total toll collected in descending order.

Query 2

For each toll plaza, list the total number of crossings, the total revenue collected, and the number of failed transactions. Group the results by plaza and include only plazas that have at least 3 crossings.

Query 3

For each FASTag account, find the total number of crossings, the total amount deducted, and the number of distinct plazas used. Include only accounts that crossed at more than 2 distinct plazas, ordered by total amount deducted in descending order.

Instructions

Diagrams

Draw an Entity Relationship diagram showing all the database tables you would create and the relationships between them (one-to-many, many-to-many, etc.). Also draw a high-level system diagram showing how the modules interact with each other.

Application

Build a console application. You do not need to connect to any database.

Data Storage

Use in-memory collections (such as HashMap, List, or equivalent in your language) to store and retrieve all records at runtime. You can hardcode a few toll plazas with lanes, toll rates, vehicles and a few FASTag accounts as initial data to start the program. All other data — toll transactions, recharges, passes, disputes, alerts — must be handled dynamically: add to the collection when created, update in place, and remove when deleted or cancelled.

OOP Concepts

Your code must clearly demonstrate the following object-oriented concepts:

- **Encapsulation** — keep fields private and expose them through methods.
- **Inheritance** — identify entities that share common properties and model them using a base class (for example, Car, Van, Bus and Truck can extend a Vehicle base class, and FASTag, Cash and Emergency lanes can extend a Lane base class).
- **Polymorphism** — handle different behaviors through a common type (for example, FASTag and Cash payments each processing a toll payment in their own way).
- **Abstraction** — hide details behind an abstract class or interface (for example, a PaymentMethod interface used by the toll service).